

Personal Statement

University of Michigan-Dearborn

As a high school student fascinated by the elegant logic of programming, I enrolled in Electrical Engineering, bowing to familial expectations, but quickly found myself constrained by the seemingly mechanical nature of the field. In my early university years, the power industry appeared to be nothing more than turning knobs and pulling levers to maintain rigid machinery. My algorithmic mindset felt entirely sidelined, trapped in a discipline that demanded mechanical compliance rather than creative problem-solving. However, my perspective fundamentally shifted during a Power System Analysis class when I realized that behind the physical infrastructure lay a beautiful, intricate order of mathematical logic and power flows. I recognized the greatness of the legacy built by previous generations, yet I longed to use computational thinking to elevate it rather than just passively operating it. From that defining moment, the grid was no longer a collection of lifeless wires, but a colossal mathematical graph waiting to be optimized.

During my time in a progressive research lab, I eagerly attempted to apply decentralized algorithms to real-world grid operations. I quickly encountered the immense historical inertia of a top-down operational mindset. This centralized legacy is a magnificent feat that has successfully ensured energy security for decades, but when faced with the unpredictability of modern renewable energy, it resorts to blunt curtailment rather than flexible optimization. I realized that my algorithmic solutions were clashing directly with a deeply ingrained philosophy of centralized control. This struggle forced me to step back and acknowledge that crafting a perfect algorithm in a laboratory is meaningless unless one respects the immense physical and historical momentum of the grid it aims to serve. I could not simply use micro-level code to patch a system that required a fundamental architectural evolution. This realization extinguished my satisfaction with being a conventional engineer and ignited a profound desire to discover a completely new operational paradigm.

Standing before a corporate environment that explicitly advised me to abandon programming and focus on fixed operational routines and bureaucratic relationship-building, I faced a defining crossroads. I saw that the influx of renewable energy was an irreversible force, acting like a tightly compressed spring against the current infrastructure. While others were content with maintaining the status quo, I knew that the traditional physical boundaries would inevitably be breached by the sheer volume of this new energy landscape. To simply conform would mean turning my back on the computational skills I had spent years cultivating. I understood that to meaningfully address the upcoming energy transition, I needed much sharper, cross-disciplinary weapons that my current environment could not provide. I realized I had to seek out an ecosystem where decentralized markets and algorithmic innovation were not just academic theories, but the foundation of reality.

I actively sought out an academic environment capable of bridging this technological divide. I soon discovered a significant challenge: very few programs successfully integrate high-level software development with high-voltage physical realities, often keeping computer science and power engineering in separate silos. The University of Michigan-Dearborn, however, uniquely resolves this friction; its progressive curriculum and research ecosystem seamlessly erase the boundaries between algorithms and physical infrastructure, offering the exact cross-disciplinary fusion I need. I am determined to immerse myself in UM-Dearborn to become a global systems architect who writes the code that will redefine resilience and stability for the power grids of the future.